Business Requirements Document

Live-Rate Jewellery Configurator, Cart & Checkout

1. Purpose

Jewellery prices are not fixed. They move with the market rate of gold, silver and platinum, which can change every few minutes. This application lets a customer configure a piece of jewellery, see exactly how its price is built up, and buy it, while guaranteeing two things:

1. **The customer is never charged a different amount from the one they last saw.**
2. **A piece with limited stock can never be sold twice**, even if two customers try to buy the last unit at the same moment.

2. Background and problem

Problem	Business impact
Metal rates move during a customer's visit	The price on screen may no longer match the cost, causing either loss to the business or a surprise charge to the customer
Customers don't trust opaque jewellery pricing	Abandoned purchases; complaints about hidden making charges or wastage
Stock is very limited (often 1–3 units per design)	Overselling leads to cancelled orders, refunds and reputational damage

3. Objectives

#	Objective	Measure of success
O1	Price every piece correctly from the live rate	Calculations match hand-worked values to the paisa
O2	Show a full, transparent price breakdown	Every component (metal, wastage, making, stones, GST) visible to the customer
O3	Hold a quoted price for a fair, fixed window	Price held for exactly 10 minutes, with a visible countdown
O4	Handle rate changes honestly	Customer told whether price went up, down, stayed the same, or the item sold out
O5	Never oversell	Proven by automated test: two simultaneous buyers of the last unit → exactly one order
O6	Never charge an unconfirmed price	Checkout blocked while any price change is unconfirmed or any item is out of stock

4. Scope

In scope

- Product listing, product page (configurator), cart page, order confirmation
- Live price calculation and breakdown
- 10-minute price lock on viewing a product and on adding to cart
- Re-pricing after lock expiry, with up / down / same / out-of-stock handling
- *Add to cart* and *Buy now*
- Concurrency-safe checkout
- Sample data (4 products), automated tests, one-command setup, documentation
- **Optional extras (delivered):** gold/silver/platinum rates changing every minute and pushed live to open pages; admin page to change rates and making-charge rules

Out of scope

- Customer accounts / login, saved addresses
- Real payment gateway (placing the order stands in for payment)
- Shipping, invoicing, returns
- Real market-rate feed (rates are simulated)
- Admin authentication beyond a shared token

5. Stakeholders and users

Role	Interest
Customer	Configure a piece, understand the price, buy with confidence
Store owner / admin	Correct pricing, no overselling, control over making-charge rules
Assessor / reviewer	Correct calculation, proven safe checkout, code quality, documentation

6. Pricing rules (business definition)

A piece's price is built from these components, in this order:

Component	Rule
Metal rate	Market rate per gram for pure (999) metal
Purity rate	Metal rate × fineness ÷ 999 (e.g. 22K = 916, 18K = 750, sterling silver = 925, Pt 950)
Metal value	Purity rate × weight of the piece
Wastage	A percentage of metal value, set per product
Making charges	Set per product, one of: ₹ per gram · % of metal value · flat amount
Stones	Sum of (stone price × count)
Subtotal	Metal + wastage + making + stones
GST	3% of subtotal

Component	Rule
Total	Subtotal + GST

Rounding: each line is rounded to the nearest paisa (half-up) once; the subtotal and total are sums of the rounded lines, so the breakdown always adds up exactly to the total shown.

7. Functional requirements

7.1 Product page

ID	Requirement	Priority
FR-1	Customer can choose **purity** (where the product offers more than one)	Must
FR-2	Customer can choose **size** / **weight**	Must
FR-3	Price breakdown updates **instantly** on every change	Must
FR-4	A **10-minute price lock** starts when the product is viewed, with a visible countdown	Must
FR-5	When the lock expires, the price is re-calculated at the live rate and the customer is told if it went up, down, stayed the same, or sold out	Must
FR-6	If the price went **up**, the customer must explicitly accept it before adding to cart or buying; the old price is shown struck through	Must
FR-7	If the price went **down**, the new price applies automatically with a note	Must
FR-8	**Add to cart** adds the item at the price shown	Must
FR-9	**Buy now** places an order immediately and handles: success, expired price, out of stock	Must
FR-10	Stock level shown ("Only 1 piece left", "Sold out")	Should

7.2 Cart page

ID	Requirement	Priority
FR-11	Each item shows the **price it was added at**	Must
FR-12	Each item has **its own countdown**	Must

ID	Requirement	Priority
FR-13	Price-change messages appear **inline on the affected item**	Must
FR-14	Price drop: applied automatically, note shown, old price struck through	Must
FR-15	Price rise: old price struck through next to new; **Accept** button; item not checkout-ready until accepted	Must
FR-16	Out of stock: item flagged; customer asked to remove it	Must
FR-17	Total makes clear **how many items still need price confirmation**	Must
FR-18	**Checkout button disabled** until every item is confirmed and in stock	Must

7.3 Checkout

ID	Requirement	Priority
FR-19	An order can **never** be placed while any item has an unconfirmed price change or is out of stock	Must
FR-20	The customer is charged **exactly** the total last shown	Must
FR-21	If two customers try to buy the last unit at the same time, **exactly one succeeds**	Must
FR-22	The unsuccessful customer is told the item has sold out	Must

7.4 Optional extras

ID	Requirement	Status
FR-23	Metal rate changes automatically every minute	Delivered
FR-24	Rate changes are pushed live to open product and cart pages	Delivered
FR-25	Admin can change making-charge and wastage rules	Delivered

8. Non-functional requirements

ID	Requirement
NFR-1	**Correctness:** no floating-point money arithmetic;

ID	Requirement
	all amounts held as whole paise
NFR-2	**Integrity:** stock can never go negative (enforced in code and by a database constraint)
NFR-3	**Responsiveness:** pages usable on phone (≈390px), tablet and desktop
NFR-4	**Setup:** runs with a single command
NFR-5	**Testability:** automated tests for price calculation, lock expiry, simultaneous orders, cart price changes
NFR-6	**Trust:** server always recalculates prices; it never accepts a price sent by the browser
NFR-7	**Accessibility:** clear status messages, keyboard-usable controls, reduced-motion respected

9. Business rules

ID	Rule
BR-1	A price lock lasts **10 minutes**.
BR-2	Viewing a product starts a lock. Adding to cart starts a **fresh** 10-minute lock at the price the customer was looking at.
BR-3	A lock captures both the metal rate and the product's pricing rules, so an admin change cannot alter a price that is already locked.
BR-4	Price decreases are applied automatically; price increases require explicit acceptance.
BR-5	An accepted price is only valid if it is exactly the price the customer was shown.
BR-6	Stock is reserved only when an order is placed, not when an item is added to cart.
BR-7	Stock is per design: one physical piece can be finished to the chosen purity and size.
BR-8	GST is a flat 3% of the subtotal (as specified in the brief).

10. Assumptions and constraints

- Rates are quoted per gram for 999-fine metal.
- Wastage is a percentage of metal value.
- In reality, Indian jewellery GST is 3% on goods and 5% on separately invoiced making charges; the brief specifies a flat 3%, which is what is implemented.
- Carts are anonymous (identified by a random ID stored in the browser).
- Quantity is 1 per add on the product page (pieces are mostly unique); the API supports larger quantities.
- Expected effort: 2–3 hours.

11. Acceptance criteria

#	Criterion	How it is verified
AC-1	Prices match hand-calculated values	pricing.test.js (hand workings in comments)
AC-2	Lock valid at 9:59, expired at 10:00	price-lock.test.js
AC-3	Two simultaneous orders for last unit → one success	concurrent-checkout.test.js (also 20 repeated rounds, 15 parallel buyers)
AC-4	Price drop auto-applied; rise needs accept; checkout blocked until confirmed	cart-price-change.test.js
AC-5	Sample data has ≥3 products, one with stones, one with a single unit	Seed data: Aarna Solitaire Ring (stones), Meher Temple Necklace (1 unit)
AC-6	Runs with one command	docker compose up --build
AC-7	README covers assumptions, running, rounding, future work	README.md

12. Sample data

Product	Metal / purity	Making rule	Stones	Stock
Aarna Solitaire Ring	Gold 22K / 18K	₹650 per gram	Diamond 0.25 ct + 2 rubies	3
Meher Temple Necklace	Gold 22K	14% of metal	—	**1**
Kaya Silver Anklet	Silver 925 / 999	₹850 flat	—	2
Ira Platinum Band	Platinum 950	₹900 per gram	—	1

13. Future enhancements

- Reserve stock when added to cart (with expiry)
- Idempotency keys so a network retry can never create a duplicate order
- Real payment gateway with stock held during payment
- Real market-rate feed and an audit trail of the rate used per order
- Separate GST rates for metal and making charges; hallmark (HUID) details
- Customer accounts and secured admin